

Chapter 3

Operators and Quantum Dynamics

Lecture 7: Time Evolution and the Schrödinger Equation

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Lecture 6 introduced operators, eigenfunctions, and the measurement postulates. This lecture first clarifies the difference between applying an operator and performing a measurement. We then introduce the Schrödinger equation as the law of time evolution, study definite-energy states, and solve the energy eigenvalue problem for a free particle.

Questions.

- Why is applying an operator to a wave function not the same as measuring an observable?
- What determines the state between two measurements?
- How do the relations $E = \hbar\omega$ and $p = \hbar k$ motivate the Schrödinger equation?
- Why does an energy eigenstate have a time-independent probability density?
- What are the energy eigenfunctions of a free particle?

3.5 The quantum postulates revisited

3.5.1 Applying an operator is not measuring an observable

An operator is a mathematical rule that maps one wave function to another:

$$\hat{A}\psi(x) = \text{another function of } x. \quad (3.31)$$

This operation is not a measurement. For example, applying $\hat{p}$ to a general state does not reveal the momentum of the particle in a single trial. Only when ψ is a momentum eigenfunction does

$$\hat{p}\psi = p\psi, \quad (3.32)$$

and the state then has the definite momentum p . For a general state, $\hat{p}\psi$ is another function, not a measured number.

The first-pass quantum postulates may be summarized as follows:

1. A state is represented by a normalized wave function $\psi(x, t)$.
2. Each observable A is represented by a linear, Hermitian operator $\hat{A}$.
3. The eigenfunctions of $\hat{A}$ form a basis in which a general state can be expanded.
4. The squared modulus of an expansion coefficient gives the probability of the corresponding eigenvalue.
5. An ideal measurement returns one eigenvalue and leaves the state in the corresponding eigenstate or eigenspace.

If a measurement at time t_* returns a , the post-measurement state supplies a new initial condition:

$$\psi(x, t_*) = \psi_a(x), \quad \hat{A}\psi_a = a\psi_a. \quad (3.33)$$

The measurement postulates describe the update at t_* . We now need a dynamical law that determines how the state changes between measurements.

3.6 Time evolution and the Schrödinger equation

3.6.1 Energy and frequency

Fundamental dynamical equations are physical postulates rather than consequences of mathematics alone. Like Newton's law, the Schrödinger equation is accepted because its predictions agree with experiment.

Planck and Einstein associated the energy of a light quantum with its frequency, and Compton scattering confirmed that a photon carries both energy and momentum. De Broglie proposed that the same relations apply to matter:

$$E = \hbar\omega, \quad p = \hbar k. \quad (3.34)$$

Consider the one-dimensional matter wave

$$\psi(x, t) = Ae^{i(kx - \omega t)}. \quad (3.35)$$

Spatial differentiation extracts its momentum:

$$\frac{\hbar}{i} \frac{\partial \psi}{\partial x} = \hbar k \psi = p \psi. \quad (3.36)$$

Time differentiation similarly extracts its energy:

$$i\hbar \frac{\partial \psi}{\partial t} = \hbar \omega \psi = E \psi. \quad (3.37)$$

Thus $i\hbar\partial_t$ is associated with energy, just as $(\hbar/i)\partial_x$ is associated with momentum.

3.6.2 The time-dependent Schrödinger equation

A general state need not have a definite energy. The number E on the right of equation (3.37) must therefore be replaced by the energy operator $\hat{H}$.

Postulate (Time-dependent Schrödinger equation). The time evolution of a nonrelativistic quantum state is governed by

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = \hat{H} \psi(x, t). \quad (3.38)$$

For one particle of mass m in a time-independent potential $V(x)$,

$$i\hbar \frac{\partial \psi(x, t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x, t)}{\partial x^2} + V(x) \psi(x, t). \quad (3.39)$$

The equation is first order in time and second order in space. Its main structural properties are:

- **Complex evolution.** The factor i is essential; a general state cannot remain purely real as it evolves.
- **Linearity.** If ψ_1 and ψ_2 are solutions for the same Hamiltonian, then $a\psi_1 + b\psi_2$ is also a solution.
- **Determinism between measurements.** Once $\hat{H}$ and $\psi(x, t_0)$ are known, the later state is determined.
- **Unitarity.** Schrödinger evolution preserves inner products and therefore preserves the normalization of the wave function.

The last property means that

$$\int_{-\infty}^{\infty} |\psi(x, t)|^2 dx = 1 \quad (3.40)$$

at every time if the initial state is normalized. This deterministic, unitary evolution applies between measurements; the ideal measurement update in equation (3.33) is a separate rule.

3.7 How to solve the Schrödinger equation

3.7.1 Specify the Hamiltonian

The Schrödinger equation cannot be solved until the Hamiltonian is known, just as Newton's equation cannot be solved until the force law is specified. For a particle in one dimension,

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x). \quad (3.41)$$

Different choices of $V(x)$ define different physical systems. For example, the harmonic-oscillator potential $V(x) = \frac{1}{2}m\omega^2 x^2$ gives

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + \frac{1}{2}m\omega^2 x^2 \psi. \quad (3.42)$$

Given an initial wave function and suitable boundary conditions, solving this equation determines the time evolution. The harmonic oscillator will be studied in detail later.

3.7.2 Solution strategies

Three complementary approaches are common:

1. **Direct analysis:** solve the differential equation exactly when possible.
2. **Structural methods:** use symmetries, conserved quantities, or ladder operators to organize the problem.
3. **Numerical methods:** discretize and solve the equation when no useful closed form is available.

The Hamiltonian, initial condition, boundary conditions, and physical scales must be identified before any of these methods can be used correctly.

3.8 Energy eigenstates and stationary states

3.8.1 Evolution of one energy eigenstate

For a time-independent Hamiltonian, first consider an initial state with definite energy. Its spatial wave function satisfies

$$\hat{H}\phi_E(x) = E\phi_E(x). \quad (3.43)$$

If $\psi(x, 0) = \phi_E(x)$, the Hamiltonian acts on this state as multiplication by E . The Schrödinger equation then reduces to

$$i\hbar \frac{\partial \psi_E(x, t)}{\partial t} = E\psi_E(x, t), \quad (3.44)$$

with solution

$$\boxed{\psi_E(x, t) = e^{-iEt/\hbar}\phi_E(x)}. \quad (3.45)$$

Equivalently, $\psi_E(t) = e^{-i\hat{H}t/\hbar}\phi_E$. Because ϕ_E is an eigenfunction of $\hat{H}$, the operator exponential reduces to the phase factor $e^{-iEt/\hbar}$.

At every fixed time,

$$\hat{H}\psi_E(x, t) = E\psi_E(x, t), \quad (3.46)$$

so the state remains an energy eigenstate and an energy measurement gives E with probability one.

3.8.2 Why it is called a stationary state

The wave function changes through an overall phase, but its probability density does not:

$$|\psi_E(x, t)|^2 = |e^{-iEt/\hbar}|^2 |\phi_E(x)|^2 = |\phi_E(x)|^2. \quad (3.47)$$

This is why a definite-energy state is called a *stationary state*. “Stationary” does not mean that the wave function is independent of time; it means that its measurable probability density is independent of time.

Energy eigenstates are special. A general initial state is a superposition of them,

$$\psi(x, 0) = \sum_n c_n \phi_n(x), \quad \hat{H}\phi_n = E_n \phi_n, \quad (3.48)$$

and linearity gives

$$\psi(x, t) = \sum_n c_n e^{-iE_n t/\hbar} \phi_n(x). \quad (3.49)$$

Each energy component acquires its own phase, so the probability density of a superposition can change with time.

3.9 Example: a free particle

For a free particle, $V(x) = 0$, and therefore

$$\hat{H} = \frac{\hat{p}^2}{2m} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}. \quad (3.50)$$

The energy eigenvalue equation is

$$\hat{H}\phi_E(x) = E\phi_E(x), \quad (3.51)$$

or

$$\phi_E''(x) + \frac{2mE}{\hbar^2} \phi_E(x) = 0. \quad (3.52)$$

Writing

$$k^2 = \frac{2mE}{\hbar^2} \quad (3.53)$$

gives the general spatial solution

$$\phi_E(x) = Ae^{ikx} + Be^{-ikx}, \quad E = \frac{\hbar^2 k^2}{2m}. \quad (3.54)$$

The two terms have momenta $p = +\hbar k$ and $p = -\hbar k$, but both have the same kinetic energy. This is an example of degeneracy: different momentum eigenstates share the same energy.

Including the stationary-state time dependence gives

$$\psi_E(x, t) = (Ae^{ikx} + Be^{-ikx}) e^{-iEt/\hbar}. \quad (3.55)$$

On the whole real line, k and E are continuous. Plane waves are not square-normalizable, so they are generalized eigenfunctions rather than ordinary normalizable states.

Questions.

1. What is the difference between calculating $\hat{A}\psi$ and measuring A ?
2. Why must the Hamiltonian and an initial state both be specified to predict time evolution?

3. Use $\hat{H}\phi_E = E\phi_E$ to verify the stationary-state solution.
4. Why is an energy eigenstate called stationary even though its wave function depends on time?
5. Why do the two free-particle states e^{ikx} and e^{-ikx} have the same energy?

Lecture 7 summary

- Applying an operator is a mathematical operation; a measurement returns an eigenvalue and updates the state.
- The Schrödinger equation determines the linear, deterministic, and unitary evolution between measurements.
- The Hamiltonian $\hat{H} = \hat{p}^2/(2m) + V(\hat{x})$ is the energy operator and determines the dynamics.
- An energy eigenfunction satisfies $\hat{H}\phi_E = E\phi_E$. Its time dependence is the phase $e^{-iEt/\hbar}$.
- An energy eigenstate is stationary because its probability density is independent of time.
- A general state is a superposition of energy eigenstates, and each component acquires its own time-dependent phase.
- Free-particle energy eigenfunctions are plane waves with $E = \hbar^2 k^2/(2m)$.

Suggested reading. Griffiths and Schroeter, *Introduction to Quantum Mechanics*, 3rd ed., Ch. 2, sections on the time-independent Schrödinger equation and stationary states. Review Lecture 6 on operators, eigenfunctions, and measurement.